

Module 8: The Central Dogma — Study Questions

DNA Structure

1. What are the three components of a nucleotide?
2. Name the four nitrogenous bases in DNA. Which are purines and which are pyrimidines?
3. State the base pairing rules for DNA. How many hydrogen bonds form between A-T? Between G-C?
4. What does it mean that DNA strands are antiparallel? Why does this matter for replication?
5. What is the difference between a gene, a chromosome, and a genome?

DNA Replication

1. Why must DNA be replicated before a cell divides?
2. What does "semi-conservative" mean in the context of DNA replication? How did Meselson and Stahl demonstrate this?
3. What is the role of helicase in DNA replication?
4. Why can DNA polymerase only build in the 5' → 3' direction? How does this create the leading and lagging strands?
5. What are Okazaki fragments? On which strand are they found? What enzyme joins them?
6. Why is an RNA primer needed to start replication?
7. If one strand of DNA reads 5'-ATCGGATC-3', what is the sequence of the complementary strand? (Include directionality.)

Transcription

1. What is transcription? Where does it occur in eukaryotic cells?
2. Which enzyme performs transcription? How does it know where to start?
3. What is the difference between the template strand and the coding strand of DNA?
4. How does RNA differ from DNA? (List at least three differences.)
5. Given the DNA template strand 3'-TACGGCAT-5', write the resulting mRNA sequence.
6. What are the three types of RNA? Briefly describe the function of each (mRNA, tRNA, rRNA).
7. What post-transcriptional modifications occur in eukaryotes? What is the purpose of the 5' cap, poly-A tail, and intron splicing?
8. What is the difference between introns and exons? How does alternative splicing increase protein diversity?

Translation

1. What is translation? Where does it occur?
2. What is a codon? How many nucleotides make up a codon?
3. How many possible codons are there? How many amino acids do they code for? What does this redundancy ("degeneracy") mean?
4. What is the start codon? What amino acid does it code for?
5. What are the three stop codons? Do they code for amino acids?
6. Describe the structure and function of tRNA. What is an anticodon?
7. Describe the three ribosome sites (A, P, E). What happens at each?
8. Walk through the steps of translation: initiation, elongation, and termination.

Application and Integration

1. Starting with the DNA sequence 3'-TACAAAGCAATT-5' (template strand):
 - Write the mRNA sequence
 - Using a codon table, determine the amino acid sequence
2. If a single base is inserted into the middle of a gene (an insertion mutation), what happens to the protein? Why is this called a "frameshift" mutation?
3. Compare and contrast DNA replication, transcription, and translation using a table with: location, enzyme/machine, template, product, and direction.
4. A mutation changes one codon from GAA to GAG. Using the codon table, determine whether this changes the amino acid. What type of mutation is this called?
5. Why is the genetic code considered "universal"? What does this suggest about the evolutionary history of life?